

Magneto-Chiral Anisotropy

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Magneto-optical effects like the Faraday effect, are intimately related to the breaking of time-reversal symmetry by the magnetic field. The breaking of space-inversion symmetry in chiral systems leads to optical activity. In both cases, the polarization of light is rotated upon propagation through the medium. Interpreting the Faraday effect as a sign of magnetically induced chirality, Pasteur was the first to investigate whether magnetic fields can discriminate between left- and right-handed chiral systems, followed by many others, but without success.

The recently discovered magnetochiral anisotropy (MChA) effect [1][2][3] can be regarded as an optical cross-effect between chirality and magnetism, or alternatively, as the consequence of simultaneously breaking time- and space symmetry. It is described by an extra term in the dielectric constant for right/left (+/-) circularly polarized light and right/left handed (*d/l*) media in the following way:

$$\varepsilon_{+/-}^{d/l}(\omega, \vec{B}, \vec{k}) = \varepsilon(\omega) \pm a^{d/l}(\omega)k \pm b(\omega)B + c^{d/l}(\omega)\vec{B} \cdot \vec{k} \quad [1]$$

where ε , a , b and c are complex ($x^d = -x^l$, Ref. [3] and references therein), $\vec{k}$ is the wavevector of the light and $\vec{B}$ is the magnetic field. The $a^{d/l}$ -term describes natural optical activity and the b -term describes magnetic optical activity. The essential features of MChA, described by the $c^{d/l}$ -term, are (i) the dependence on the relative orientation of $\vec{B}$ and $\vec{k}$, (ii) the dependence on the handedness of the chiral medium (enantioselectivity) and, (iii) the *independence* of the polarization state of the light. The relative strength of MChA in optical absorption with unpolarized light is given by

$$g_{MChA} \equiv 2 \frac{A(\vec{B} \uparrow \uparrow \vec{k}) - A(\vec{B} \uparrow \downarrow \vec{k})}{A(\vec{B} \uparrow \uparrow \vec{k}) + A(\vec{B} \uparrow \downarrow \vec{k})} \approx \frac{2c'' \vec{B} \cdot \vec{k}}{\varepsilon''} \quad [2]$$

where the A 's are the optical extinction coefficients for the respective configurations. The most convenient experimental setup to measure g_{MChA} consists of an unpolarized or a linearly polarized light source in combination with an alternating magnetic field, as described in Refs. [1] and [3]. In this way, MChA measurements can be done in almost any spectral range. Figure 1 shows a MChA absorption spectrum of the two enantiomers (mirror images) of the chiral Cr(III)tris-oxalato complex (*d/l*-Cr(ox)₃³⁻) in solution. MChA has also been measured in luminescence for chiral europium complexes [1] and for chiral nickel sulphate crystals in absorption [3]. The real part of the MChA c -term has recently been measured by an interferometric technique for several molecular liquids and crystals [2] [4]. A molecular, quantitative theory has been developed for MChA [5], which shows that both combined electric dipole-magnetic dipole and electric dipole-electric quadrupole transitions may contribute to this effect in molecular solutions, in contrast to natural circular dichroism (NCD), where the quadrupole contributions vanish in solution.

Enantioselective photochemistry with circularly polarized light (CPL) is well-established [6]. It is based on NCD, i.e. a difference in absorption coefficient of the two enantiomers for CPL. The Cr(ox)₃³⁻ complex has been extensively studied in this context as it is unstable in solution and spontaneously dissociates and re-associates. In equilibrium, one has a racemic mixture (equal concentrations of right- and left handed complex). The dissociation is accelerated by the absorption of light, so under CPL irradiation, the more absorbing enantiomer will dissociate more often. This leads to an excess of the less absorbing enantiomer, the handedness of which depends on the handedness of the CPL [7].

Similarly, due to MChA, one should expect enantioselectivity in photochemical reactions when one-sidedly irradiating a racemic solution of Cr(ox)₃³⁻ with unpolarized light (UPL) in a magnetic field parallel to the irradiation direction. The handedness of the excess will be determined by the relative orientation of light and field. Its magnitude

is given by $ee_{\text{UPL+B}} = g_{\text{MChA}}/2$. This conjecture has been recently confirmed by experiments [8]. For a magnetic field parallel to the irradiation direction a strictly linear relation between excess and field is observed, giving $ee/B = 1.0 \cdot 10^{-5} \text{ T}^{-1}$, which is close to the calculated excess. For perpendicular fields no significant ee is observed. These results are in full agreement with the properties of MChA and constitute the irrefutable proof that photochemistry with unpolarized light in magnetic fields can have enantioselectivity.

In view of the above it would seem interesting to study MChA in the X-ray spectral range. The measurement does not require a special polarization state and alternating the magnetic field gives good sensitivity and resolution. As the effect can be present in isotropic solutions, it can be used to study molecular properties that are not accessible to X-ray NCD.

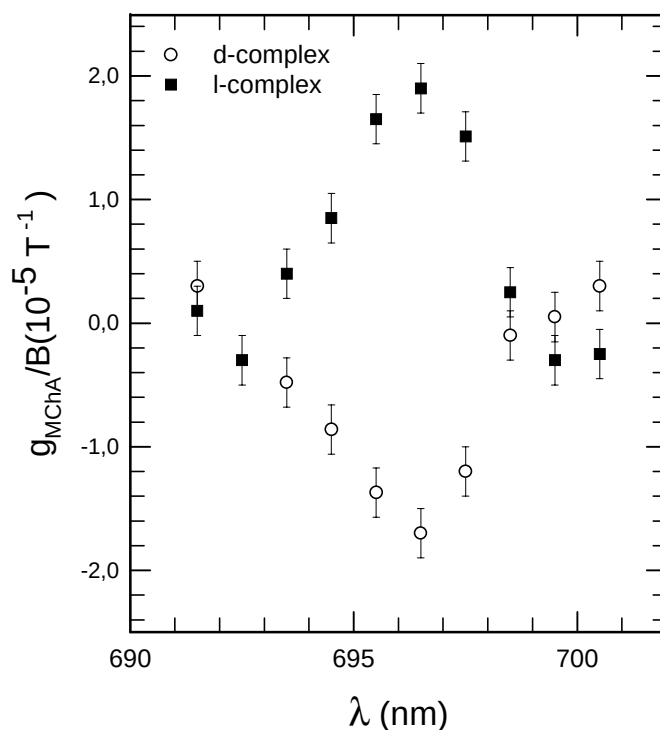


Figure 1 MChA spectra of the two $\text{Cr}(\text{ox})_3^{3-}$ enantiomers in dimethylsulfoxide

References

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